artdmx-bridge32

Art-Net to DMX512 Gateway

User Manual

Firmware version 2.3.0-web-dmx

Document revision 1.0 — June 2026

Table of Contents

1. Safety and intended use

Important. This device interfaces with low-voltage control wiring (DMX512, RS-485, GPIO). It does not replace qualified electrical installation for mains-powered fixtures. Install and operate in accordance with local regulations.

The artdmx-bridge32 converts **Art-Net** lighting data received over WiFi into **DMX512** output for professional lighting control. It is intended for fixed installations, portable rigs, and bench testing where Art-Net is the primary control source.

- Operate only with the recommended ESP32 supply voltage and properly rated RS-485 transceivers.
- Do not expose the device to moisture unless your enclosure is rated for the environment.
- Use shielded or twisted-pair DMX cable suitable for RS-485.
- Terminate long DMX runs according to fixture manufacturer guidance.

2. Product overview

artdmx-bridge32 is an ESP32-based gateway that:

- Receives **Art-Net** universes over WiFi
- Outputs **DMX1** on the primary RS-485 port (always active)
- Optionally outputs **DMX2** on a second RS-485 port with channel filtering
- Optionally accepts **wired DMX input** on port 2 as fallback when Art-Net is idle
- Provides a built-in **web dashboard**, **configuration pages**, and **DMX test tools**
- Stores settings in **non-volatile flash (NVS)**

> **Note.** Port 2 is used either as **DMX2 output** or as **DMX1 wired input**, never both at the same time. Changing this mode requires a reboot.

Key features

Feature	Description
Web dashboard	Live status, traffic indicators, rolling log, setup and test-mode banners
Web configuration	WiFi, hostname, Art-Net, DMX timing, debug logging, factory reset
Configuration backup	Download/upload editable `artdmx-bridge32.conf` text files
DMX test page	16 manual sliders; Art-Net ignored while test mode is active
mDNS	Access via `http://<hostname>.local`
Setup access point	Automatic AP if WiFi connection fails within 5 seconds
ArduinoOTA	Wireless firmware updates from the Arduino IDE

3. Requirements and specifications

Hardware requirements

Component	Specification
Controller	ESP32 module (e.g. ESP32-WROOM-32, ESP32-WROOM-DA)
RS-485	MAX485 or compatible — one for DMX1, optional second for port 2
LEDs (optional)	GPIO 13, 14, 22 with series resistors (330 Ω–1 kΩ)
Reset input (optional)	GPIO 15 to GND, hold 3 s for factory reset
Power	Stable 3.3 V / 5 V per your ESP32 board specification

Software / network requirements

Item	Requirement
WiFi network	2.4 GHz WPA/WPA2 (or open AP for initial setup)
Lighting software	Art-Net sender (universe must match device configuration)
Web browser	Modern browser for dashboard and configuration
Arduino IDE (OTA)	For wireless firmware updates; ESP32 board package 2.x or 3.x

Default operating parameters

Parameter	Factory default
Hostname	`artdmx-bridge32`
Art-Net universe	0
Art-Net timeout	15000 ms
DMX refresh interval	30 ms
DMX packet size	Full 512-channel packets enabled
DMX2 output	Disabled
DMX1 wired input	Disabled
Channel filter	Channels 1–512
Serial Monitor	115200 baud

4. Hardware installation

4.1 DMX1 output (primary port)

ESP32 GPIO	MAX485 pin	Function
GPIO 17	DI	UART TX → DMX data
GPIO 16	RO	UART RX
GPIO 4	DE + /RE	Driver enable (tie DE and /RE together)
GND	GND	Common ground
—	A / B	DMX+ / DMX− to fixtures

4.2 Second port (DMX2 output or DMX1 input)

ESP32 GPIO	MAX485 pin	Function
GPIO 19	DI	TX for DMX2 output
GPIO 18	RO	RX for wired DMX1 input
GPIO 21	DE + /RE	Driver enable
GND	GND	Common ground
—	A / B	DMX+ / DMX− line

4.3 Status and traffic LEDs

GPIO	Function
GPIO 13	DMX1 output activity (~150 ms flash per packet)
GPIO 14	DMX2 output or DMX1 input activity
GPIO 22	WiFi / OTA / DMX test status (see Section 12)
GPIO 15	Factory reset — hold LOW 3 seconds

5. First-time setup

5.1 Flash firmware

1. Install the ESP32 board package and required libraries (`esp_dmx` 4.x, `ArtnetWifi`).
2. Copy `secrets.h.example` to `secrets.h` and enter default WiFi and OTA credentials.
3. Open `artdmx-bridge32.ino` in the Arduino IDE, select your board and port, and upload.
4. On ESP32 Arduino core 3.3.x, an `esp_dmx` library patch may be required for compilation and DMX1 wired input — see project README.

5.2 Verify boot

Open the Serial Monitor at **115200 baud**. A successful boot shows:

```
artdmx-bridge32 2.3.0-web-dmx starting...
Copyright (C) 2026 Richard Smetana
Licensed under GNU GPL v3 or later
DMX test mode off (default after boot)
```

5.3 Connect to WiFi

The device uses WiFi credentials from NVS (after first web save) or from `secrets.h` on first boot. If connection fails within **5 seconds**, a setup access point starts:

Item	Value
Setup SSID	`<hostname>-setup` (default: `artdmx-bridge32-setup`)
Web UI	Gateway IP shown in Serial log and dashboard banner

Connect your computer or phone to the setup SSID, open the device IP in a browser, and enter your WiFi credentials on the **Configuration** page. Save and reboot if prompted.

6. Network access

6.1 Access URLs

Method	URL / address	When to use
mDNS	`http://<hostname>.local`	Normal operation on LAN (default: `http://artdmx-bridge32.local`)
IP address	`http://<device-ip>`	When mDNS is unavailable
Setup AP	Gateway IP from Serial log	When WiFi is not configured or unreachable

6.2 DHCP hostname

The configured **hostname** is sent to your router as the DHCP client name. It should appear in the router client list as `artdmx-bridge32` (or your custom name), not `esp32-XXXXXX` . Change hostname on the Configuration page and reboot if your router caches old leases.

6.3 WiFi reliability

The firmware reconnects automatically if the link is lost. Link health is checked every **2 seconds** (valid IP and AP association). The status LED single-blinks while connecting.

6.4 Web login (optional)

If a **Web password** is set on the Configuration page, all web pages require HTTP basic authentication:

Field	Value
Username	Device hostname (default: `artdmx-bridge32`)
Password	Web password from Configuration

Leave Web password empty to disable authentication. WiFi and OTA passwords are separate.

7. Web interface

All pages include a navigation bar (**Dashboard · Config · DMX Test**), logo, hostname header, and a footer with copyright and GPL license link.

7.1 Dashboard (`/`)

Refreshes every **500 ms**.

Area	Description
Setup banner	Orange banner in AP mode with setup SSID and URL
DMX test banner	Purple banner when DMX test mode is active
DMX traffic	Web indicators for DMX1 out, DMX2 out, DMX1 in — active ~2 s after last packet
Status grid	Hostname, URL, IP, RSSI, uptime, heap, OTA, Art-Net, DMX test, universe, refresh, port modes, filter
Serial log	Last 32 log lines; auto-scrolls unless you scroll up

7.2 Configuration (`/config`)

Button / action	Description
Save	Write settings to flash
Save and Reboot	Save settings and restart device
Reboot	Restart without saving changes
Download .conf	Export all settings as text file
Upload / Upload and Reboot	Import settings from file or pasted text
Factory reset	Erase all settings (confirmation required)

7.3 DMX Test (`/dmx-test`)

See Section 9 for full DMX test mode documentation.

8. Configuration reference

Settings are stored in NVS under namespace `artdmx-bridge32`. After the first web save, values persist across reboots.

Setting	Default	Notes
WiFi SSID / password	From secrets.h	Reboot recommended after change
Hostname	artdmx-bridge32	mDNS, OTA, DHCP name, web username (max 31 chars)
Web password	(empty)	Enables HTTP basic auth when set
OTA password	From secrets.h	Empty = OTA without password
Art-Net universe	0	Must match lighting software
Art-Net timeout	15000 ms	Idle time before wired input fallback
DMX refresh interval	30 ms	Re-send rate for hold-style dimmers
Send full 512-channel packet	On	Off sends only active channel range
Enable DMX2 output	Off	Filtered Art-Net mirror on port 2; reboot required
Enable DMX1 wired input	Off	Disables DMX2; uses port 2 as input; reboot required
Channel filter start / end	1 / 512	Port 2 channel range — see Section 11
Art-Net console log	Off	Off / matched / ignored / all universes
Debug log channel start / end	1 / 4	Channels shown in log lines
Log every Nth packet	1	Reduce log volume (e.g. 10, 100)
Log only on value change	Off	Skip identical consecutive frames

> **Reboot required** for WiFi, hostname, DMX2 output enable, and DMX1 wired input mode changes to take full effect.

9. DMX test mode

The DMX Test page allows manual control of DMX1 for bench testing and commissioning.

Control	Description
DMX test mode	When enabled, Art-Net is ignored and slider values drive DMX1. Dashboard shows purple banner. GPIO 22 status LED breathes.
Send DMX only when slider moved	Default ON — DMX sends only when you move a slider. OFF = continuous refresh at configured interval.
Update sliders from Art-Net	When test mode is OFF, mirrors incoming Art-Net on page sliders (display only, 500 ms poll).
Start channel	First of 16 consecutive channels (1–497)
16 sliders	Range 0–255; loaded from last Art-Net frame when page opens
Buffer dumps	Slider range view and full 512-channel buffer; Refresh button updates full dump

> **Runtime only.** DMX test mode and its checkboxes reset to OFF after every power cycle. They are not stored in NVS.

10. Configuration backup and restore

Export and import all permanent settings as a plain-text file `artdmx-bridge32.conf`.

10.1 Download

Configuration page → **Download .conf**, or `GET /api/config/download`.

10.2 Upload

Select a file or paste text → **Upload** or **Upload and Reboot**.

10.3 File format

- One setting per line: `key=value`
- Lines starting with `#` are comments
- Booleans: `true` or `false`
- Quote values with spaces: `wifi_pass="my secret"`
- All keys from the downloaded file are required on upload

> **Validation.** Syntax errors, unknown keys, duplicate keys, missing keys, or illegal values cause upload to be **rejected**. Existing settings are **not changed** on failure.

11. Channel filter (port 2)

The channel filter applies to DMX2 output and DMX1 wired input on port 2.

DMX2 output mode

- Only channels inside the filter range are sent on DMX2 (others zeroed in that packet).
- DMX2 transmits **only when a value in the filter range changes** — unchanged refresh cycles skip DMX2.
- DMX1 always receives the full Art-Net frame.

Typical use: Modern fixtures on DMX1; legacy fixtures prone to flicker on DMX2 with a narrow filter (e.g. channels 31–63).

DMX1 wired input mode

When Art-Net is idle beyond the timeout, wired DMX on port 2 is read. Only channels inside the filter range are forwarded to DMX1 output.

12. LED indicators

GPIO 22 — Status LED

Pattern	Meaning
Steady on	WiFi connected
Single blink	Connecting to WiFi
Double blink	Setup access point active
Triple blink	OTA firmware update in progress
Breathing (PWM fade)	DMX test mode active

GPIO 13 / 14 — Traffic LEDs

Flash for approximately **150 ms** on each DMX packet. Disabled during OTA. Web dashboard traffic dots follow packets for ~2 seconds (refresh-only cycles do not activate them).

13. Art-Net operation

1. Configure lighting software to send Art-Net to the device IP address.
2. Set the Art-Net **universe** in software to match the device configuration (default universe 0).
3. DMX1 output updates from matched universes immediately.
4. If DMX2 output is enabled, filtered channels update only on change within the filter range.
5. If DMX1 wired input is enabled and Art-Net stops for longer than the timeout, wired input takes over on DMX1 output.
6. While **DMX test mode** is active, Art-Net is ignored.
7. During **OTA updates**, Art-Net and all DMX I/O are disabled.

The device short name in Art-Net follows the configured hostname.

14. Firmware updates (OTA)

1. Ensure the device is on WiFi and OTA is ready (dashboard shows **OTA: Ready**).
2. In Arduino IDE: **Tools** → **Port** → select network port (e.g. `artdmx-bridge32 at ...`).
3. Upload sketch. Enter OTA password when prompted (if configured).

During OTA: DMX outputs blackout first, Art-Net stops, GPIO 13/14 traffic LEDs off, GPIO 22 triple-blinks until complete.

15. Factory reset

Method	Procedure
Web UI	Configuration → Danger zone → Factory reset → confirm
GPIO 15	Hold pin LOW to GND for 3 seconds

Erases all NVS settings and reboots with defaults from `config.h` and `secrets.h`. DMX test mode settings are always reset on boot regardless.

16. Serial Monitor and logging

Connect USB serial at **115200 baud**. The log mirrors the dashboard (WiFi events, config changes, Art-Net debug, OTA, DMX test on/off). OTA progress percentage appears on Serial only.

Enable Art-Net console logging on the Configuration page to debug universes and channel values. Use **Log every Nth packet** and **Log only on value change** to control volume.

17. Troubleshooting

Symptom	Recommended action
No DMX output	Check wiring, DE pin, Art-Net universe, sender IP/subnet
Web UI unreachable	Try device IP; join setup AP; check web password
Router shows esp32-XXXXXX	Set hostname, reboot, renew DHCP lease
DMX2 never updates	Enable DMX2 output; verify filter range; only changes in range trigger send
Legacy fixtures flicker on DMX2	Narrow channel filter; DMX2 sends only on change
Wired input ignored	Enable DMX1 input, reboot, wait for Art-Net timeout
Traffic dots stay Idle	Confirm universe match; dots follow packets not refresh-only
Config upload rejected	Read error message; fix syntax and required keys
DMX test stuck on	Reboot — test mode is not persistent
OTA fails	Same network; correct OTA password
Compile error uart.c	Apply esp_dmx patch for ESP32 core 3.3.x (see README)
Reboot loop with DMX input	Apply esp_dmx UART2 context fix

18. Technical support information

Item	Value
Product name	artdmx-bridge32
Firmware version	2.3.0-web-dmx
Manual revision	1.0 — June 2026
Copyright	© 2026 Richard Smetana
License	GNU General Public License v3 or later (see LICENSE)
SPDX identifier	GPL-3.0-or-later
Configuration namespace	artdmx-bridge32 (NVS)
Default hostname	artdmx-bridge32
Default web URL	http://artdmx-bridge32.local
Setup AP SSID	artdmx-bridge32-setup
Serial baud rate	115200

For developer documentation, library patches, API details, and source code layout, refer to the project **README.md** supplied with the firmware source.

18.1 License

This firmware is free software licensed under the **GNU General Public License version 3** (or later). You may redistribute and modify it under the terms of the GPL. The complete license text is in the **LICENSE** file supplied with the source code. There is **no warranty**.

Source files are marked `SPDX-License-Identifier: GPL-3.0-or-later`. If you distribute modified firmware, you must provide corresponding source code and document your changes, as required by the GPL.